MIL-T-53019(ME)
4 January 1983

MILITARY SPECIFICATION

TRUCK, FORKLIFT, ELECTRIC, FRONT/SIDE LOADING

6000/6000 POUND CAPACITY

TYPE E

COMMERCIAL

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a front/side loading electric forklift truck having a capacity of 6000 pounds.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, (see 6.2) the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-S-43926	- Suit, Chemical Protective.
MIL-G-43976	- Glove Set, Chemical Protective.
MIL-F-43987	- Footwear Cover, Chemical Protective (Overboots).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3930

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MIL-C-46168	- Coating, Aliphatic Polyurethane, Low. Reflective, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-M-51282	- Mask, Chemical-Biological, Field, M17A1.
MIL-H-51291	- Hook, Chemical-Biological Mask: M6A2.

STANDARDS

MILITARY

MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications, standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

29 CFR, Chapter XVII, Part 1910.
Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

TECHNICAL INFORMATION PACKAGE (TIP)

(Application for copies should be address to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HM, Fort Belvoir, VA 22060.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B 56.1 - Safety Standard for Powered Industrical Trucks.
MR 11.3 - Load Handling Symbols for Powered Industrial Trucks.
MH 11.4 - Forks and Fork Carriers for Powered Industrial Trucks.
ANSI/UL 583 - Standard for Electric Battery Powered Industrial Trucks.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Standards.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The forklift shall be front/side loading capable of pivoting the load not less than 90 °, sit-down rider with a load capacity of 6000 pounds on a 24 inch load center at full lift height in both the front and side carry position. It shall have a 36-volt electric powered system. The forklift shall be equipped with a battery, electric drive motor, battery discharge indicator, a roller mast, and as specified herein. The forklift shall be the latest model of the manufacturer's basic commercial product and shall have been in production, marketed, and in use for a minimum of one year preceding the solicitation for procurement. The forklift shall be equipped with standard instruments, components and accessories normally required for the safe and effective operation of the truck. The forklift shall be complete with all components that are listed as standard on the contractor's literature for the basic product, whether stipulated herein or not, together with such accessories and options as may be specified herein. The forklift shall be equipped with all other components and parts not specifically mentioned but necessary to provide functional machine and shall conform, in quality, to that normally provided to the commercial industry (see 6.5). Forklift requirements shall be as specified herein.

3.2 First article. The contractor shall furnish a forklift(s) for examination and testing within the time frame specified (see 6.2) to prove that forklifts comply to the requirements of this document. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government.

3.3 Safety. The forklift shall conform to the requirements of ANSI B 56.1, ANSI/UL 583, (type E), the Occupational Safety and Health Standards requirements of 29 CFR, Chapter XVII, Part 1910, and all pertinent safety standards in effect at the time of solicitation.

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3.4 Environment.

3.4.1 Operating temperature requirements. The forklift shall be equipped with all standard and optional equipment to enable it to operate normally at full capacity in any ambient temperature from 0° F to +110° F. When specified (see 6.2) the forklift shall be equipped with a kit to enable it to operate at -50° F.

3.4.2 Nuclear-Biological-Chemical (NBC) protection. The forklift shall be operable by personnel wearing the following complete protective ensemble:

a. Chemical-biological mask, MIL-M-51282.
b. Chemical-biological hood, MIL-H-51291.
c. Chemical protective suit, MIL-S-43926.
d. Chemical protective glove set, MIL-G-43976.
e. Chemical protective footwear, MIL-F-43987.

The operator shall be capable of actuating all controls (i.e., levers, switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the forklift. The forklift shall complete two cycles of operation as specified in 4.4.11. The cycles shall be completed in 10 minutes or less.

3.5 Human factors. The characteristics of the forklift shall provide for operation and maintenance in the environments described in paragraph 3.4 by the clothed 5th to 95th percentile male or female personnel (as defined in SAE J833) who are suitably clothed for these environments. The design shall also incorporate the characteristics described in SAE J920A, J899, J297, J898, J298, J209, J925, J1042, J614B and J231.

3.5.1 Visibility. The forklift shall be designed so that the 5th percentile female through the 95th percentile male can view an area adjacent to the right side of the forklift when it is carrying a 20 foot long, 43 inch wide, 43 inch high load placed on the deck in the side carry position. This area shall extend the length of the load and includes a view which is not less than 5 feet beyond the right lateral edge of the load. If a device is used to provide the view and the device is not within the plan outline of the forklift it shall be easily collapsible so that it will not be damaged by a 5 mph collision with a stationary hard object such as a door jamb. The forklift, using the viewing device, shall be capable of performing the test specified in paragraph 4.4.17.

3.6 Sound level. The sound level at the operator's position shall not exceed 85dBA as specified for Category D of MIL-STD-1474. It shall be tested in accordance with paragraph 5.1.2 of MIL-STD-1474.

3.7 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication, except where sealed permanent lubrication is provided. The types of lubricants shall be selected from, and in accordance with, SAE J754. All initial fills of lubricants shall be limited to the types approved by the component manufacturers. Grease lube fittings shall conform to SAE J534.

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3.8 Maintainability. Provision shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system components, battery, wearing parts of lift and tilt mechanism brakes and components wear parts of steering assembly, tires, wheels, lights, and horn. If hand access openings are required to perform maintenance, the edges of each opening shall be smooth and the opening shall be provided with a removable or hinged cover. Dimensions of access openings shall be in accordance with SAE J925 for bare hand. There shall be no interference with the servicing or drawing of oils and lubricants to or from any assembly or components by frame members or other obstructions, and without requiring the use of flexible connections all reservoirs with removable drain plugs shall provide the complete drainage.

3.9 Maintenance operations. Each operation in the following maintenance operations list shall be accomplished by one person in not more than the time specified, using common tools and any special tools, furnished with the truck:

Maintenance Operation	(Time in Minutes)
a. Remove and replace drive motor brushes.	90
b. Remove and replace hoist and tilt motor brushes.	90
c. Remove and replace steer motor brushes.	90
d. Remove and replace all contactor tips.	90
e. Remove and replace all filters, screens and strainers in hydraulic system.	105
f. Remove and replace brake shoes, springs, cylinder and any other brake component. Bleed and refill master cylinder (two people).	45

3.10 Electrical equipment.

3.10.1 Enclosures and shields. All electrical components including wiring shall be so located or protected by adequate means to prevent contact with or the entrance of water.

3.10.2 Safety cutout system. Provisions shall be made to disrupt power to the traction motor when the operator leaves his seat.

3.10.3 Speed controller. A solid state speed controller shall be provided. The controller shall provide stepless or combination step and stepless acceleration through the complete speed range in both the forward and reverse directions and shall have automatic electrical or mechanical interlocks to prevent starting except in low speed. It shall provide automatic means to limit the plugging when the controller is reversed with the forklift in motion. All contactors shall be designed for operation on 36 volts dc and shall have easily replaceable tips of copper or silver alloy composition. Replacement of contactor tips shall be accomplished by use of handtools only.

3.10.4 Electric motors. The traction and pump motors shall be industrial type with a high starting torque. The insulation for motors shall be class B, F, or H as defined NEMA standards.

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3.10.5 Battery. The forklift shall be furnished with a 36 volt battery having a minimum capacity of 1275 AH at the 6 hour rate. A spare battery, of the same voltage and capacity, shall be furnished with each forklift.

3.10.6 Battery discharge indicator. Each forklift shall be furnished with a battery discharge indicator containing a lift interrupt. The lift circuit shall be interrupted when the battery has been discharged 80 percent.

3.10.7 Electromatic Pulse (EMP) protection. The forklift shall be protected against high altitude electromagnetic pulses. The contractor shall install protective devices as specified by the Government. Selection of the protective devices will require Government evaluation of the forklift. The contractor shall furnish a forklift, with all associated wiring, diagrams and electrical schematics, detailed to the component level (i.e. resistors, capacitors, filters, transistors, SCR's) with the value of all components given, to the Government for evaluation.

3.11 Battery charger. When specified (see 6.2) a battery charger shall be furnished with each forklift. The charger shall be capable of operating on w3 phase, 208/230/460 volt, 47-63 hertz ac input. The charger shall be furnished with a charging receptacle that is compatible with the battery connector. The charger shall be capable of recharging a 1275 AH battery, which has been discharged 80 percent, to its full capacity in 8 hours or less.

3.12 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of all load handling functions. Hydraulic fittings on pumps, valves, and cylinders shall be the O-ring straight thread type as specified in SAE J514 or 4-bolt flanged type as specified in SAE J518. The system shall provide for lowering of the rated load at the rate specified in 3.21.2. A pipe tap or straight thread O-ring boss test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. All pressure hose fittings shall be either 37-degree flare or 4-bolt split flange as specified in SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum relief valve setting. The hose installation shall not cause the hose to bend beyond the minimum bend radius specified in SAE J517. Pressure tube fittings shall be 37 degree flare or O-ring boss conforming to SAE J518. A system filter shall be furnished in either the suction or return line. Hose reels shall be mounted such that they are within the plan outline of the forklift.

3.13 Uprights and carriage assembly. Uprights shall be the telescopic roller type. Main load rollers shall be the permanently lubricated for life type. Chains shall be installed in such a manner so that they cannot be displaced from the roller until they have been raised a minimum of 3/8 inch. The lift shall he hydraulically actuated, with the upper portion automatically lifting as the load is raised above the "free lift" position. Positive means shall be provided to prevent over-travel of the carriage or mast channel in both the upper and lower positions.

3.13.1 Forks and fork carrier. The forks and fork carrier shall conform to ANSI B56.1 and MH 11.4. The fork carrier shall be free of obstructions to allow the forks to slide across the entire width of the carrier. A cutout

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shall be provided in the fork carrier so that the lower hook of the fork will fit through the slot and each fork shall be removable without removing the load backrest. Unless otherwise specified (see 6.2) forks shall be 22 inches long. Fork thickness shall not exceed 1/2 inch at the tips and end taper shall be not less than 14 inches long. The maximum fork spread shall be 37 inches measured between the outer edges of the forks.

3.13.2 Load backrest. A metal backrest, removable without disturbing any other component, shall be provided. There shall be no protruding bolts or appendages beyond the side plane of the load backrest. In conjunction with the fork and hanger design, it shall provide a vertical rear guard at least 48 inches high, measured from the load carrying surface of the forks, and shall be at least equal to the width of the carrier.

3.14 Pivot and shift assemblies. A pivot and shift assembly shall be provided to allow the load to be pivoted 90 degrees in either one or two directions. It shall also allow the load to be side shifted. Pivot and shift shall be accomplished hydraulically. Pivot shall be minimum of 90 degrees in each direction used. With the forks in the side carrying mode, traverse shall be sufficient to allow the front face of the forks to extend not less than 4 inches beyond the plan outline of the forklift. With the forks in the front carry position side shift shall be a minimum of 22 inches. With the forks in the side carry position side shift be a minimum of 44 inches. The pivot and shift assemblies shall be capable of withstanding the tests specified in 4.4.8 and 4.4.9 without failure or permanent deformation.

3.15 Wheels and tires.

3.15.1 Wheels. The forklift shall have steel, malleable iron, or cast iron wheels, equipped with tapered roller bearings.

3.15.2 Tires. The tires shall be new and unused, solid or cushion style, and industrial type. Front tires shall have commercial, nondirectional type tread; rear tires may have either smooth rib or nondirectional type tread. Composition shall be rubber or polyurethane.

3.16 Steering. Power steering shall be furnished.

3.17 Brakes. Hydraulically actuated, self ajusting brakes shall be provided. The reservior shall be filled with silicone brake fluid conforming to the requirements of MIL-B-46176.

3.18 Controls and instrumentation. All controls and instrumentation shall be provided with position markings. Load handling controls shall be marked in accordance with ANSI MH 11.3. The controls shall be operable and the instruments shall be viewable by the 5th to 95th percentile male and female, according to the dimensions of SAE J833a.

3.18.1 Key switch. A key-operated on-off switch shall be furnished. Two keys shall be furnished with each forklift. All keys shall be made to fit each and every forklift provided under the contract.

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3.18.2 Hour meter. An electrically operated hour meter which registers the total number of hours of operation of the traction and pump motors shall be provided.

3.19 Lights. Each forklift shall be equipped with four sealed beam flood lamps, two mounted in the front and two mounted in the rear. An automotive type combination stoplight and tail-lamp shall also be furnished and mounted at the rear.

3.20 Operator's overhead guard. A removable overhead guard in accordance with ANSI B56.1 shall be provided. The guard shall be removable with hand tools. Guard height, with the mast tilted rearward, measured from the ground to the highest point of the guard shall not exceed 93 inches.

3.21 Performance.

3.21.1 Lifting speed. Lift speed with rated load on the forks shall be not less than 30 feet per minute when measured over the entire distance from ground level to maximum fork height.

3.21.2 Lowering speed. Speed of lowering of the unloaded forks shall be not less than 40 feet per minute average over the entire distance from maximum fork height to the ground. Speed of lowering the forks with rated load on them shall be not more than 80 feet per minute over the entire distance from maximum fork height to ground level.

3.21.3 Speed. While carrying the rated capacity load on a level surface the forklift shall be capable of attaining a speed of not less than 5.0 mph in both the forward and reverse direction.

3.21.4 Slope ascension, forward direction. The forklift shall be capable of ascending a 15 percent grade on a dry concrete surface with and without rated load. The forklift shall be able to accelerate from a dead stop on the slope with and without rated load.

3.21.5 Mast tilt. When unloaded the forklift mast shall have a forward tilt of 3 degrees, plus or minus 1/2 degree, and a rearward tilt of 4 degrees minimum.

3.21.6 Collapsed mast height. The collapsed mast height of the unloaded forklift measured from the ground to the top of the upright shall not exceed 93 inches.

3.21.7 Maximum fork height. The maximum fork height of the forklift bearing rated capacity load with the uprights fully extended, measured from the ground to the load carrying surface of the forks at the rated load center, shall be not less than 144 inches.

3.21.8 Free lift height. With the forks in the front and side carry position the mast shall have a free lift of not less than 5 inches above the height of the side carry deck, without increasing the collapsed mast height.

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3.21.9 Aisle width. The forklift, with a 20 foot long, 43 inch wide, 43 inch high, 6000 pound capacity load on the forks shall be capable of operating in a 89 inch wide aisle to insure the loading and unloading of stacked containers etc., with the close quarters area.

3.21.10 Stability. Truck shall conform to the stability requirements as follows:

Operating as Counterbalanced Front Loader		Operating as Counterbalanced Front/Side Loader	
Test (Stability)	Platform Slope Value %	Test (Stability)	Platform Slope Valve %
Long. Stacking	4	Long. Stacking	4
Long. Travel	24	Long. Travel	24
Lat. Stacking	6	Lat. Stacking (RH)	6
		Lat. Stacking (LH)	6
Lat. Travel	ANSI B56.1	Lat. Travel	ANSI B56.1

3.21.11 Productivity. The forklift, with fully charged battery, shall operate for 30 cycles as specified in 4.4.11 without degrading the travel speed below 2 miles per hour or the lift speed below 20 feet per minute with the rated load on the forks.

3.22 Treating and painting. The portions of the forklift normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, forest green. If another color is specified (see 6.2), the above system shall be overtreated with MIL-C-46168 of the desired color.

3.23 Identification marking. Each forklift shall be identified with an identification plate conforming to MIL-P-514, type 1, style 1, composition C, of type 1, grade A, class 1 materiel. The plate shall be located in a visible and permanment location. The identifying data shall be stamped in letters in accordance with MIL-P-514. The plate shall be securely attached to the forklift with screws, bolts, rivets, or by welding. The plate shall be furnished and mounted by the contractor.

3.23.1 Forklift marking. Markings shall be painted black. Characters shall be block-type capitals and Arabic numerals. Location, information and character size shall be as follows:

Information	Example	Location	Minimum Character Size (Ht.)
Agency	US	On each side and rear	1-1/2 inches
Registration Number	10D315	On each side and rear	1-1/2 inches
Capacity	6000 lb. cap	On each side	2 inches
Safety	No smoking	On rear of mast	1 inch
	No riders	On rear of mast	1 inch

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3.24 Technical manuals. One copy of the contractor's parts, maintenance, and operator's manual(s) shall be provided with each forklift at time of shipment. The manual(s) shall be placed in a transparent, heat sealed bag. The bag shall be attached to the forklift.

3.25 Technical information package (TIP). The manufacturer shall complete the TIP to depict the forklift and its components as it will be furnished. Notation shall be made of all components listed in the TIP which have not been in field use for a minimum of 1 year, and shall include identification of the component or part previously used. The manufacturer shall certify, by signing the TIP, that the item to be furnished is the same as those being supplied for commercial use, except for modifications permitted in 3.1 of this specification. The TIP shall be submitted with all bids and proposals.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Notwithstanding the requirements for any Government inspection and test contained in specifications applicable to this contract, except where specialized inspections or tests are specified for performance solely by the Government, the contractor shall perform or have performed the inspections and test required to substantiate that the supplies and services provided under the contract conform to the drawings, specifications and contract requirements listed herein, including, if applicable, the technical requirements for the manufacturer's part numbers specified herein.

4.2 Classification of inspections and tests. Inspections and tests shall be classified as follows:

a. First article test (see 4.3).
b. Production run-in (see 4.5).
c. Inspection of packaging (see 4.7).

4.3 First article demonstration. Prior to delivery, the manufacturer shall make two or more forklifts available (see 6.2), selected at random by the Government from forklifts being produced from production tooling for first article evaluation and demonstration. The truck and its components shall be compared with this document, the contractor's completed TIP (see 3.25), and published literature, calculations and test data, to verify compliance with this specification. The forklift(s) shall be subjected to a demonstration as specified herein. Permanent deformation, overheating, malfunction, failure to meet performance requirements, failure to complete any test, or leakage of lubricants shall be cause for rejection of the forklift(s). If such deficiencies do arise, they shall be corrected and the affected test shall be repeated.

4.3.1 Schedule of examinations. The following examinations shall be conducted as a part of the first article to assure conformance to Section 3 requirements.

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Defect	Requirement Paragraph
Safety provisions not as specified.	3.3
Human factors not as specified.	3.5
Lubrication provisions not as specified.	3.7
Electrical equipment not as specified.	3.10 (inclusive)
Hydraulic system not as specified.	3.12
Uprights and carriage assembly not as specified.	3.13
Wheels and tires not as specified.	3.15
Steering not as specified.	3.16
Brakes not as specified.	3.17
Controls and instrumentation not as specified.	3.18 (inclusive)
Lights not as specified.	3.19
Operator's overhead guard not as specified.	3.20
Treating and painting not as specified.	3.22
Identification marking not as specified.	3.23

4.4 Tests.

4.4.1 Test conditions. Tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to commencement of the test shall be performed during the test.

4.4.2 Rated load. Unless otherwise specified herein, a 48 inch cube of the specified rated load shall be used. Rated load is defined as the forklift capacity specified in 3.1. The tolerance for weight shall be plus 0 percent and minus 1 percent and for dimensions shall be plus or minus 1 percent.

4.4.3 Maximum fork height, collapsed mast height, free-lift height. With the rated load on the forks and the mast in the true vertical position measure and record collapsed mast height, free lift height and maximum fork height. Nonconformance to 3.21.6, 3.21.7, or 3.21.8 shall be cause for rejection of the forklift.

4.4.4 Lifting speed and lowering speed. Measure and record the distance from the floor to the top surface of the forks with the forks in the fully lowered position. Subtract this figure from maximum fork height. Using the resultant figure raise the rated load from the floor to maximum fork height and record the time in seconds. Record time in seconds to lower loaded forks at maximum speed from maximum height to fully lowered position. The raising and lowering shall be performed three times and averaged. Remove land and raise empty forks to maximum height. Record time in seconds required to lower forks to fully lowered position. Nonconformance to 3.21.1 or 3.21.2 shall be cause for rejection of the forklift.

4.4.5 Travel speed. Operate the forklift with the rated load in the front carry position. The forklift shall attain the specified speed in both the forward and reverse directions. The forklift shall be driven the distance necessary to attain maximum speed prior to entering a measured distance of not less than 22 feet. The forklift shall be driven over the measured course three times in each direction. The speed shall be the average of the three

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runs in each direction. Inability of the forklift to attain minimum speed as specified in 3.21.3 shall be cause for rejection of the forklift.

4.4.6 Slope ascension. With the rated load on the forks and in the front carry postion, drive the forklift on to a ramp or slope having a 15 percent grade. Bring the forklift to a complete stop halfway up the ramp or slope. Start from a dead stop and proceed up the ramp. Repeat the procedure without the load. Nonconformance to 3.21.4 shall be cause for rejection of the forklift.

4.4.7 Tilt. Place the unloaded forklift on a flat level surface. With the forks in the front carry position raise them to an elevation of 2 feet and place the mast in the true vertical position. Tilt the mast to the full forward position and record the angle of tilt. Tilt the mast to the full rearward position and record the angle of tilt. Nonconformance to 3.21.5 shall be cause for rejection of the forklift.

4.4.8 Sideshift. Operate the sideshift with the forks in the front carry position for 50 cycles at full speed from one extreme to the other with a 6000 pound load on the forks. Next operate the sideshift with the forks in the side carry position for 50 cycles at full speed from one extreme to the other with a 6000 pound load on the forks. During the last 25 cycles of each test, the control shall be released abruptly when the forks are at approximately mid-position between the left and right extremes, and then re-engaged. Upon completion of the tests the pivot and shift assembly shall be examined for defects. Nonconformance to the requirements of 3.14 shall be cause for rejection of the forklift.

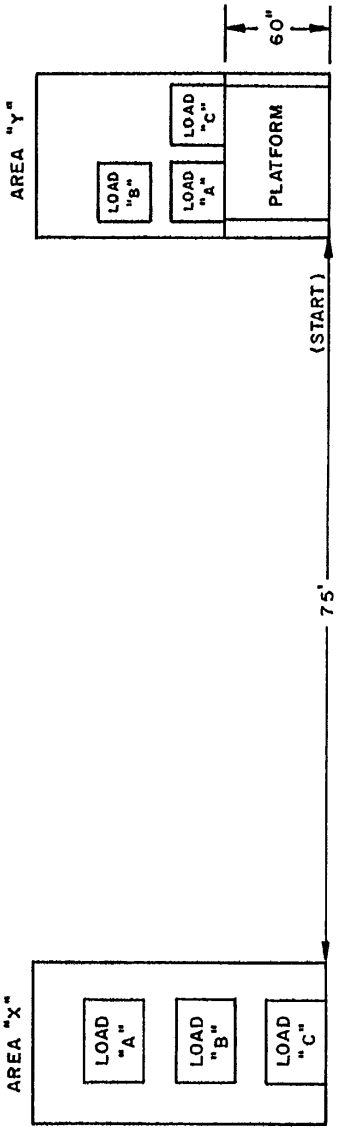
4.4.9 Pivot. Operate the pivot for 100 cycles at full speed from one extreme to the other with a 6000 pound load on the fork. During the last 50 cycles, the control shall be released abruptly when the forks are at approximately in the mid-position between the left and right extremes, and then re-engaged. Upon completion of the 100 cycles, the pivot and shift assembly shall be examined for defects. Nonconformance to the requirements of 3.14 shall be cause for rejection of the forklift.

4.4.10 Aisle width. Place a 20-foot long, 43-inch wide, 43-inch high 6000 pound capacity load on the forks and pivot to the side carry position. Place the load on the deck. Drive the forklift in both the forward and reverse direction in an aisle not greater than 89 inches wide. It shall be driven in each direction 3 times. The forklift shall be driven a distance of 75 feet in each direction. Nonconformance to 3.21.9 shall be cause for rejection of the forklift.

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4.4.11 Productivity. The forklift shall perform the following operation. The course shall be set up in accordance with figure 1. With the forks in the front carry position travel to area "X". Pivot the forks to the side carry position and retrieve load "A". Lower the load and travel to area "Y". Pivot the load to the front and deposit it on the platform. Return to area "X" and retrieve load "B" using the same sequence of operation. Return to area "Y" and deposit load "B" on top of load "A". Return to area "X" and retrieve load "C" using the same sequence of operation. Return to area "Y" and place load "C" beside load "A". Back out and lower the forks to the floor. Raise the forks and retrieve load "C" and return it to area "X". Deposit the load and return to area "Y". Retrieve load "B" and deposit it on top of load "C" at area "X". Return to area "Y" and retrieve load "A". Deposit the load on top of load "B" at area "X". This constitutes one cycle of the test. The forklift shall perform 30 cycles continuously using only one battery. The battery shall not be charged during the test. The forklift shall only be allowed to stop to change operators. Upon completion of the test travel speed and lift speed with rated load shall be checked. Nonconformance to the requirements of 3.21.11 shall be cause for rejection.

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X-3786A

FIGURE 1. Power consumption course.

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4.4.12 Environmental conditions. The contractor shall certify that the forklift complies with the temperature requirements of 3.4. If a certification is not furnished, the forklift shall be demonstrated as follows to prove conformance with 3.4:

a. Service the forklift with applicable lubricants.
b. Place the forklift in a temperature-controlled chamber.
c. Obtain a stabalized temperature of 0° F in the battery electrolyte.
d. Energize the forklift and operate the steering, lift, tilt, pivot, and sideshift controls for 10 minutes.
e. Engage the directional control and move the forklift forward and reverse within the confines of the test chamber. This operation shall be performed continuously for 20 minutes.
f. All operations shall be performed with a 6000 pound load on the forks.

The above demonstration shall be repeated at 110° F. Nonconformance to 3.4 shall be cause for rejection.

4.4.13 Sound level. The contractor shall certify that the forklift complies with the sound level requirements of 3.6. If a certification is not furnished the sound level of the forklift shall be checked to prove conformance to 3.6. Nonconformance to 3.6 shall be cause for rejection.

4.4.14 Maintainability. Using common hand tools and any special tool furnished with the forklift, perform the following. Examine the forklift for accessibility of major assemblies and position of drains with respect to accessibility. Using the tools, demonstrate that major assemblies are accessible for repair and maintenance. Demonstrate both the accessibility of drains and the paths of discharge of lubricants by opening the drainage controls. Perform and time each operation on the maintenance operations list (see 3.9). Nonconformance to the requirements of 3.8 or failure to perform the maintenance operations in allotted time specified in 3.9 shall be cause for rejection of the forklift.

4.4.15 Forklift operation. The forklift, carrying the rated capacity load, shall be operated for a minimum of 500 cycles over a surface paved with concrete, asphalt, macadem or equivalent. Each cycle shall consist of the following:

a. Raise the rated load to the forward carry position and tilt the mast rearward.
b. Travel forward a sufficient distance to allow the forklift to operate for a minimum of 50 feet at maximum speed.
c. Slow the forklift to a safe speed and negotiate a 90 degree turn. Stop the forklift.
d. Shift the load to the extreme left and then across to the extreme right.
e. Pivot the load 90 degrees and then return to the forward position. Shift the load back to the center of the forklift.
f. Raise the load to maximum lift height at maximum lift speed.
g. Lower the load, at maximum lowering speed, to the carry position.

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h. Tilt the mast to the full forward position and return it to the full rearward position.
i. Drive the forklift back to the initial starting point. This constitutes one cycle of operation.
j. 250 cycles shall be performed making a right hand 90 degree turn, and 250 cycles shall be performed making a left hand 90 degree turn.
k. The lights, horn and key switch shall be operated after each 50 cycles of operation.

Inability of the forklift to lift the rated load, overheating, leakage of fluid, malfunction, failure of any component, or failure to complete the operation shall be cause for rejection of the forklift. Normal battery changes are permitted, when required, to conduct the test.

4.4.16 Stability. Perform "forward stacking", “forward travel”, "lateral stacking", and "lateral travel" tilting platform tests in accordance with the appendix. Inability to attain the stabilities specified in 3.21.10 before overturning shall constitute failure of the test.

4.4.17 Visibility. Prepare a test course with 89 inch wide aisles. One aisle shall be 75 feet long with a 19 foot 90 degree intersecting aisle of the same length. Barriers 87 inches high, shall be placed continuously along both sides of the aisles. Place a 20-foot long, 43-inch wide, 43-inch high load on the forks in the side carry position on the deck of the forklift. Two operators conforming to the 5th percentile female and 95 percentile male shall be used. The operators shall use only the viewing device while performing this test. Drive the forklift forward to the intersecting aisle. Make a 90 degree right or left turn into the aisle and proceed to the end of the aisle. Return the forklift to the starting point by driving it in reverse. This cycle shall be performed 10 times using alternate operators and alternate right and left turns at the intersecting aisle. Failure of the forklift to negotiate the course or striking the barrier more than 2 times shall be cause for rejection.

4.5 Production run-in. Each forklift, carrying a rated load, shall be subjected to a production run-in and servicing as specified in 4.5.1 thru 4.5.3. Inability to complete the run-in, overheating, failure of any components, malfunction of any control, or evidence of leakage of fluids shall be cause for rejection of the forklift by the Government until such deficiencies are corrected.

4.5.1 Travel. Demonstrate starting. ascending and descending a 15 percent ramp, steering, and high speed travel. Operate all lights, horn, and the brake system not less than two times during the demonstration.

4.5.2 Mast assembly. Demonstrate maximum lifting and lowering speed, forward and rearward tilt, sideshift, and pivot of the load. Each operation shall be performed at least three times with the rated load on the forks.

4.5.3 Servicing and adjustment. Prior to delivery, the manufactuer shall service and adjust each forklift for customer operational use, including at least the following:

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a. Adjustment of mast assembly.
b. Adjustment of solid state control modules.
c. Adjustment of brake system.
d. Adjustment of steering.
e. Continuity check of the electrical system.
f. Filling and charging of the battery.

4.6 First article demonstration report. The contractor shall prepare and submit the first article demonstration report to the contracting officer for approval within the time frame specified (see 6.2). The report shall include a detailed description of the examination and demonstration performed. All deficiencies encountered and actions taken to correct the deficiencies shall be included.

4.7 Inspection of packaging. The preservation, packing, and marking of forklifts shall be inspected for conformance to the requirements of appendix B, MIL-STD-162.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A or B. Preservation, packing, and marking shall be in accordance with MIL-STD-162, level A or B as applicable, for type I or II, as specified (see 6.2).

5.1.2 Commercial. Each complete forklift and all items issued with it shall be suitably preserved and packed for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coating, volatile corrosion inhibitor, or desiccated packs. Items requiring protection from physical damage to surfaces or which are fragile by nature shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the forklift in operation. Marking shall reflect as a minimum, the item name, national stock number, quantity, contract number and appropriate address markings on each unpacked item or shipping container. The marking shall be applied with a waterproof material and shall provide a definite contrast.

6. NOTES

6.1 Intended use. Forklifts described herein are intended for stacking and unstacking, and moving long loads and pallet loads of missiles and ammunition in and around storage magazines. Forklifts are intended for operation over paved and other hard surfaces for short distances.

6.2 Ordering data. Acquisition documents should specify the following:

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a. Title and date of this purchase description.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1)
c. Number of forklifts required for first article demonstration (see 4.3).
d. Time frame for submitting first article demonstration report (see 4.6).
e. Degree of preservation and packing required (see 5.1).
f. Type of preservation and packing required (see 5.1.1).
g. Time frame for submitting first article (see 3.2).
h. Color specified if other then forest green (see 3.22).
i. When a cold weather kit is required for temperature below 0 ° F (see 3.4).
j. When a battery charger is required (see 3.11).
k. Fork length required (see 3.13.1)

6.3 Technical information package. The TIP may be obtained from US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HM, Fort Belvoir, VA 22060.

6.4 Data requirements. The contracting officer should include such data as technical publications, instructional materials, illustrated parts lists, and maintenance and operation manuals to be furnished with each forklift.

6.5 Material Deterioration and Prevention Control (MADPAC). The materials and processes used in the construction of the forklift shall reflect the requirements of DARCOM Regulation 702-24 (MADPAC). Assurance shall be provided by the contractor or vendor that the material for protective finishes used in the forklift shall provide maximum protection against various forms of material deterioration in the applicable operating and storage environments to which the forklift may be exposed (see 3.1).

Custodian:
Army – ME

Preparing activity:
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APPENDIX

TRUCK, FORKLIFT, ELECTRIC, FRONT/SIDE LOADING

6000/6000 POUND CAPACITY

TYPE E

COMMERCIAL

10. SCOPE

The test method contained within this appendix is to determine whether trucks procured under this specification conform to the requirements set forth.

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APPENDIX

TEST METHOD

LONGITUDINAL AND LATERAL STABILITY

1. Test Course:

   (a) Tilt table.

2. Test Apparatus:

   (a) Clinometer
   (b) Plumb bob
   (c) Tape measure

3. Test Procedure:

   A Counter Balanced Front Loading Operation

   (a) Longitudinal Stability - Stacking

   (1) Place the truck on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform. (Ref. Figure A-1).
   (2) When placing the rated load on the forks, the center of mass of the load shall be on the longitudinal centerline of the mast which shall be on the longitudinal centerline of the truck. For trucks with tilting uprights with the load supported by the forks approximately 12 inches above the platform, and in a normal and unrestrained manner with the stationary mast being vertical with the level platform, attach a plumb line at the theoretical intersection of the vertical face and top surfaces of the forks to establish a reference point. (A transit or other method may be used instead of a plumb line to establish vertical.)
   (3) With the rated load elevated to maximum height, and for trucks with tilting upright, the test platform shall be tipped to a 4 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

   (b) Longitudinal Stability - Traveling

   (1) Place the truck on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform. (Ref. Figure A-1).
   (2) When placing the rated load on the forks, the center of mass of the load shall be on the longitudinal centerline of the mast which shall be on the longitudinal centerline of the truck. The mast shall be at full rearward tilt. The load

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APPENDIX

is then elevated until the point of intersection of the front vertical face and top surface of the forks is 12 inches from the floor.

(3) With the load elevated as described, tip the test platform to a 24 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by wheel chocks.

(c) Lateral Stability - Stacking

(1) Placement of Truck on Tilt Platform

a. Place truck with an articulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of tilting platform. Steering wheels shall be positioned parallel to the tilting axis of the platform (see Figure A-2).

b. Place truck, with a nonarticulated steer axle, on an initially level platform in a position that a line connecting the center of right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steer wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-3).

(2) Place the rated load on the forks, the center of mass of load shall be on the longitudinal centerline of the mast. Shift the mast and/or the load to the position of least lateral stability. Elevate the load approximately 12 inches above the platform and position mast in true vertical position using level and plumb bob, or equivalent, to a marked reference point on the platform.

(3) Elevate the load to maximum height, and reposition mast to the reference point as required. The test platform shall be tipped a 6 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

(4) Repeat a. and b. using left load wheel in lieu of right load wheel.

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APPENDIX

(d) Lateral Stability - Traveling (No Load)

(1) Placement of Truck on Tilt Platform

a. Place empty truck with a articulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of the tilting platform. Steering wheels shall be positioned parallel to the tilting axis of the platform (see Figure A-4).
b. Place empty truck, with a non-articulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steering wheels shall be positioned parallel to the tilting axis of the platform (see Figure A-5).
c. Position the steering wheel nearest the tilting axis parallel to the tilting axis of the platform.

(2) Position the mast to full rearward tilt and elevate the forks until the point of intersection of the front vertical face and top surfaces of the forks are 12 inches from the floor.
(3) With the forks elevated as described, tip the test platform to the minimum slope specified in ANSI B56.1. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.
(4) Repeat a. and b. using left load wheel in lieu of right load wheel.

B Counterbalanced Front/Side Loading Operation

(a) Longitudinal Stability - Stacking

(1) Place the truck on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform (Ref. Figure A-6).
(2) Place a 6000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast with the forks in the front carry mode. The load shall be pivoted to the position of least longitudinal stability. Elevate the load approximately 12 inches above platform and

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position using level and plumb bob, or equivalent to a marked reference point on the platform.

(3) With the load elevated to maximum height and with mast repositioned to the reference point as required, the test platform shall be tipped to a 4 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

(b) Longitudinal Stability - Traveling

(1) Place the truck on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform. (Ref. Figure A-7).

(2) Place a 6000 pound load on the forks, and pivot the load 90° to either the right or left. Position load such that the center of mass of the load is on the intersection of the centerline of the truck and mast. Position the mast at full rearward tilt. Elevate load to a position where the intersection of the vertical front face and top surface of forks are 12 inches from the floor.

(3) Tip the test platform to a 24 percent slope. During the test there should be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by wheel chocks.

(c) Lateral Stability - Stacking (Right Hand)

(1) Placement of Truck

a. Place truck, with an articulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of tilting platform. Steering wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-8).

b. Place truck, with a nonarticulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steering wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-9).

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(2) Place a 6000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast which shall be on the longitudinal centerline of the truck with the forks in the front carry mode. Pivot and side shift the load to the position of least lateral stability. Elevate the load approximately 12 inches above the platform and position mast in true vertical position using level and plumb bob, or equivalent.
(3) With the load elevated to maximum height, and the mast repositioned to true vertical, the test platform shall be tipped to a 6 percent slope. During the test there should be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.
(4) Repeat test (c) using left load wheel in lieu of right load wheel.

(d) Lateral Stability - Traveling

(1) Placement of Truck

a. Place empty truck with an articulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of the tilting of the platform. Steering wheels shall be positioned such that they are parallel to the tilt axis of the platform (see Figure A-10).
b. Place empty truck, with a nonarticulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steering wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-11).

(2) Place a 6000 pound load on the forks and pivot the load 90 degrees from the longitudinal centerline of the chassis and shift to the position of least stability. If the truck has a tilting mast, the mast shall be either vertical or at full rearward tilt, whichever is less stable. Elevate the forks until the point of intersection of the front vertical face and top surface of the forks are 12 inches from the floor.
(3) Tip the platform to the minimum slope specified in ANSI B56.1. During the test there shall be no operator on the truck. Maintain the truck position on the platform by

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APPENDIX

brakes or other similar means, but not by means of wheel chocks.

(4) Repeat test using right load wheel in lieu of left load wheel.

(e) Lateral Stability - Stacking (Left Hand)

(1) Placement of Truck

a. Place empty truck with an articulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of the tilting of platform. Position steering wheels parallel to the tilting axis of the platform (see Figure A-12).

b. Place empty truck, with a non-articulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Position steering wheels parallel to the tilting axis of the platform (see Figure A-13).

(2) Place a 6000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast with the forks in the front carry mode. The load shall be tilted toward the platform tilt axis, shifted and pivoted to the position of least stability.

(3) With the load elevated to maximum height, the test platform should be tipped to a 6 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or similar means, but not by means of wheel chocks.

(4) Repeat test using right load wheel in lieu of left.

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APPENDIX

IN ALL FIGURES (A-1 THROUGH A-13

CONCERNING STABILITY TESTING

The following identifications apply:

A-A = Test Platform Tilt Axis
B-B = Axis about Which Truck Upsets
C-C = Parallel to Tilting Axis of Test Platform
L-L = Truck Longitudinal Centerline
I = Points of Intersection
WT = Wheel Tread
WB = Wheelbase

Tilting platform is tilted "up" about axis by raising opposite edge of platform.

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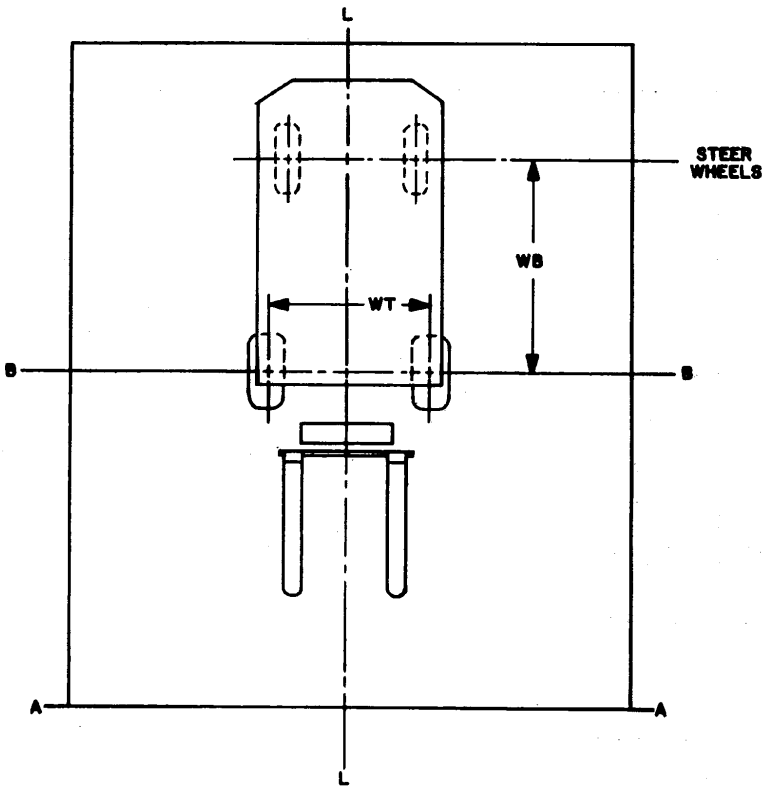
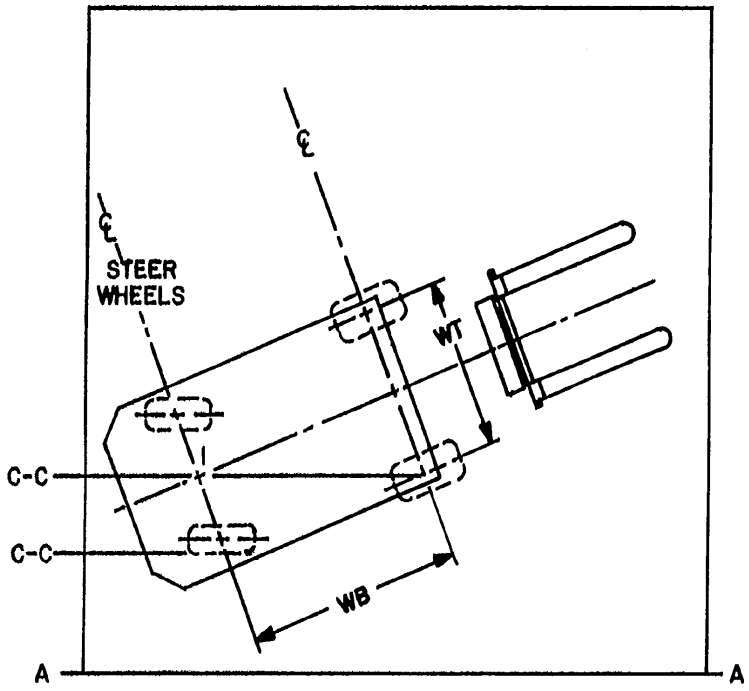


FIGURE A-1. Longitudinal stability (Counterbalanced front loading operation.)

X-3983A

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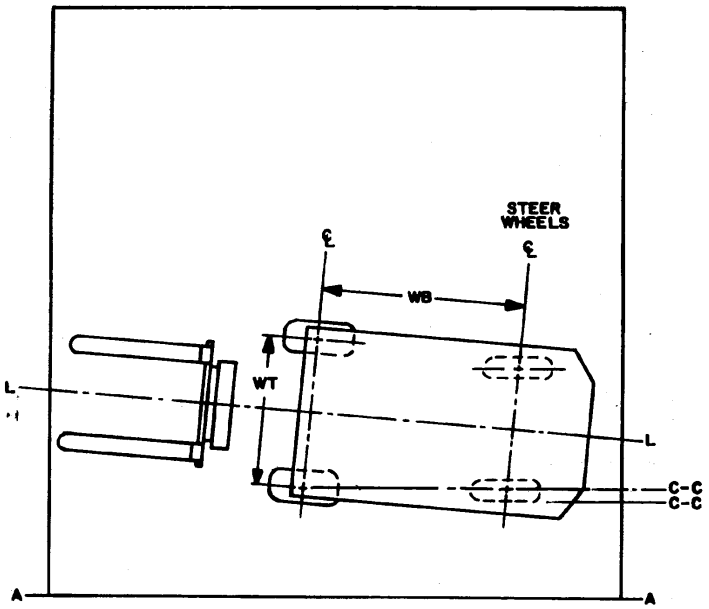


NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

FIGURE A-2. Lateral stability-stacking. X-3984A
(articulated steer axle)
(counter balance front loading operation).

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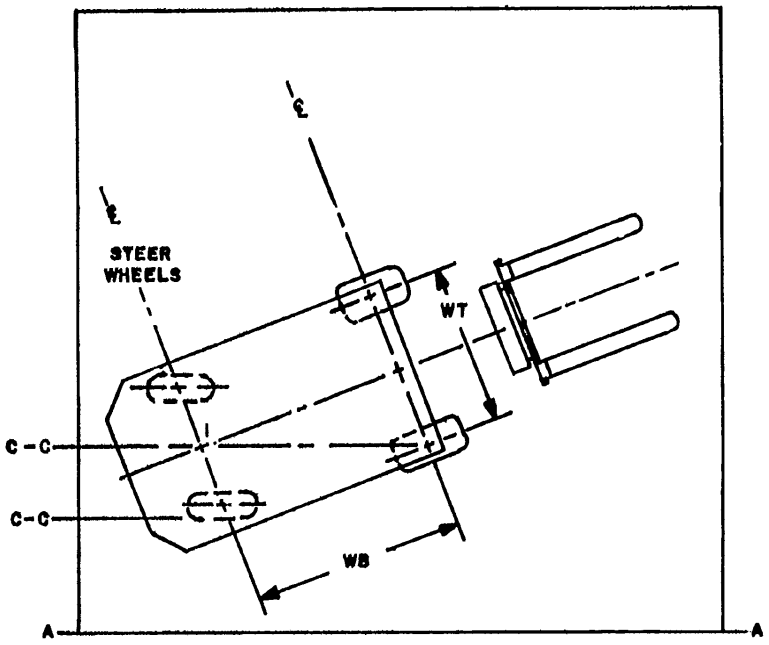
NOTE:
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FIGURE A- 3. Lateral stability-stacking (non-articulated steer axle) (counterbalanced front loading operation).

X-3985A

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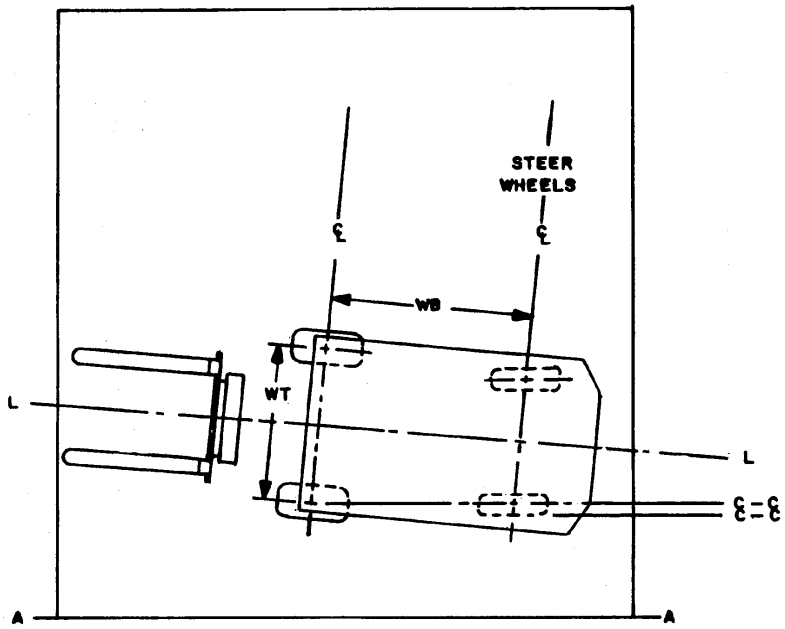


NOTE:
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FIGURE A-4. Lateral stability-traveling. (articulated steer axle) (counterbalanced front loading operation). X-3986A

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NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

FIGURE A-5. Lateral stability - traveling. X-3987A
(non-articulated steer axle)
(counterbalanced front loading operation).

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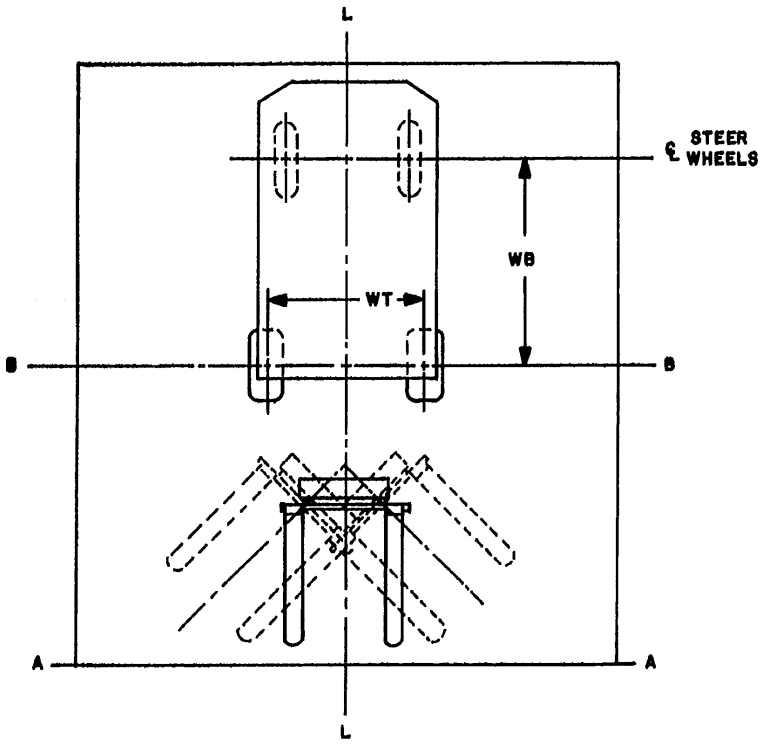


FIGURE A-6. Longitudinal stability - stacking. (Counterbalanced front/side loading operation.) X-3988A

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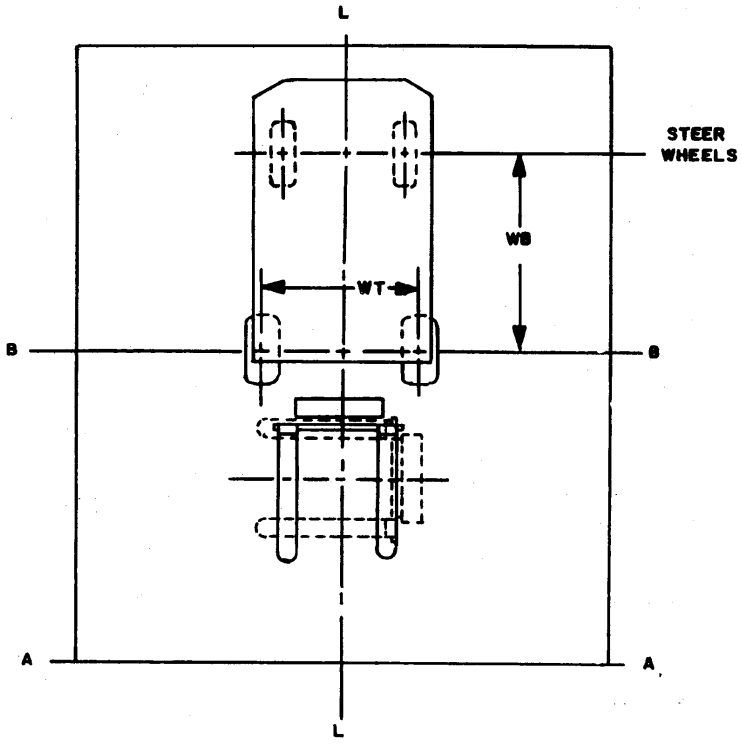
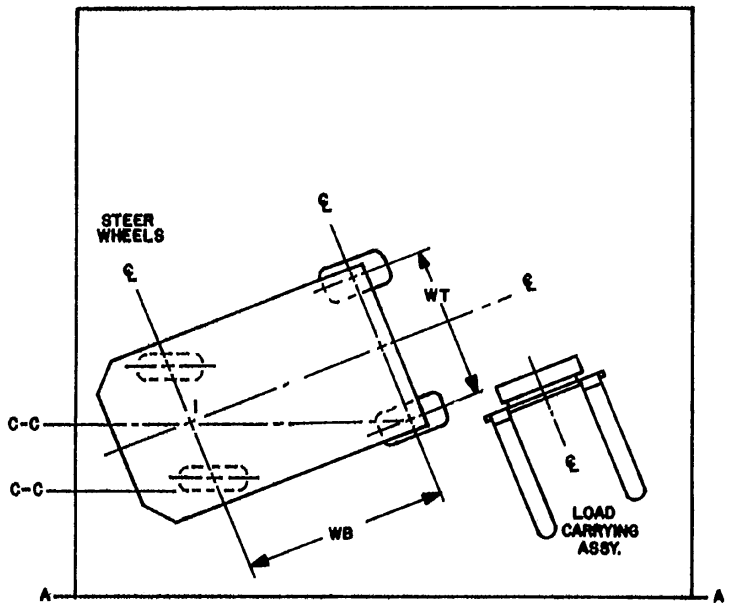


FIGURE A-7. Longitudinal stability-traveling. X-3989A

(Counterbalanced front / side loading operation.)

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NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

FIGURE A-8. Lateral stability-stacking, (articulated steer axle) (counterbalanced front/side loading operation). X-3990A

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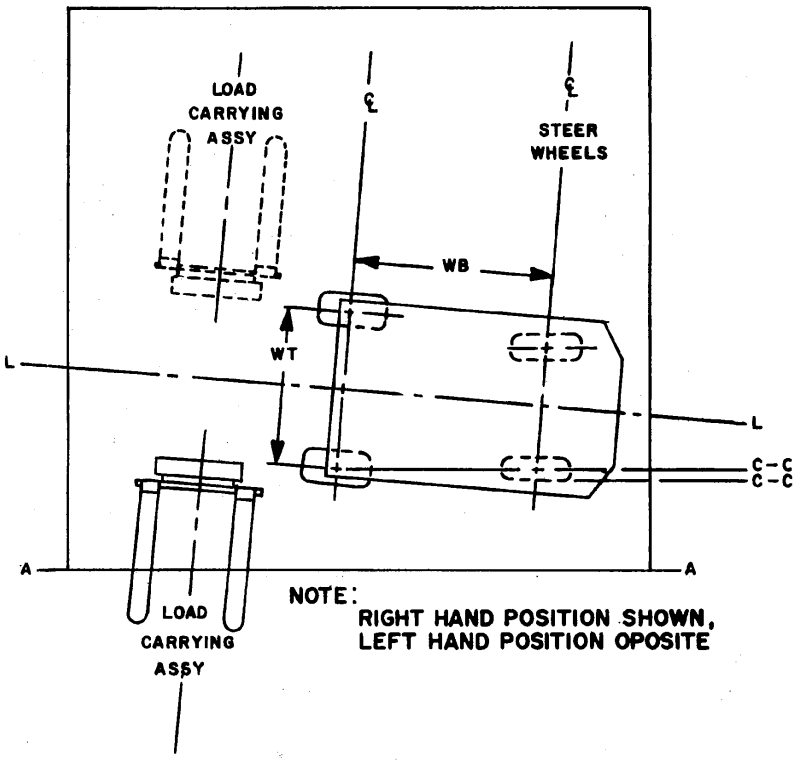
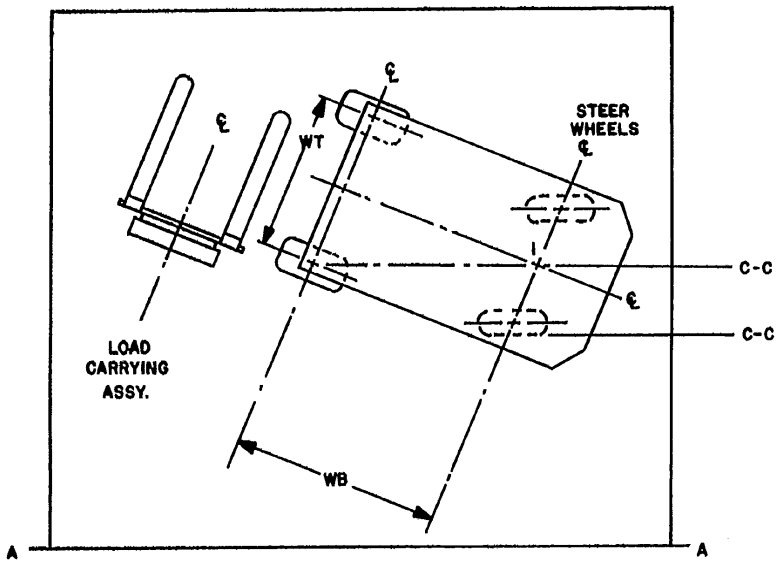


FIGURE A-9. Lateral stability-stacking. (non articulated steer axle) (counterbalanced front/side loading operation). X-3991A

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NOTE:

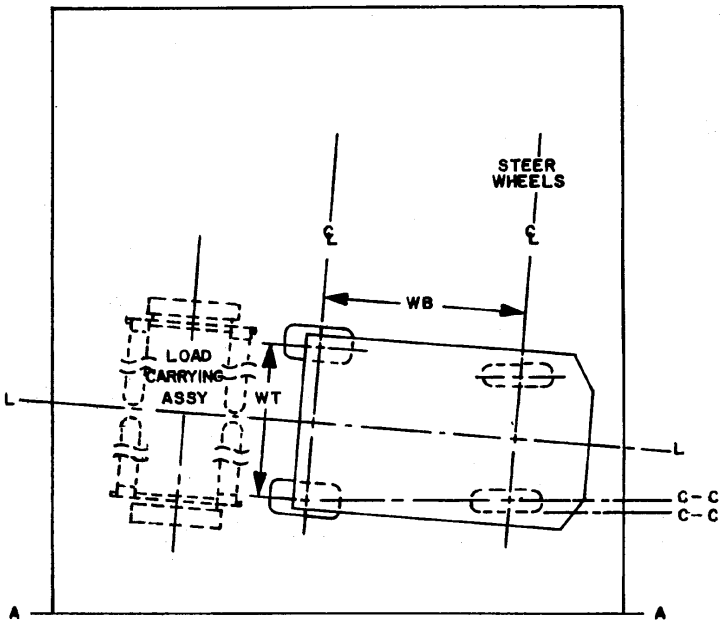
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

FIGURE A-10. Lateral stability - traveling, (articulated steer axle) (counterbalanced front/side loading operation.

X-3992A

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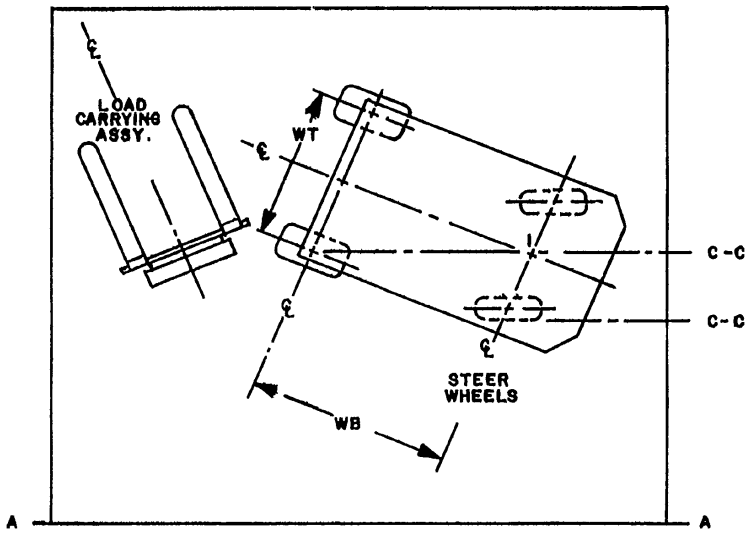
NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

FIGURE A-11. Lateral stability-traveling.
(non articulated steer axle)
(counterbalanced front/side loading operation).

X-3993A

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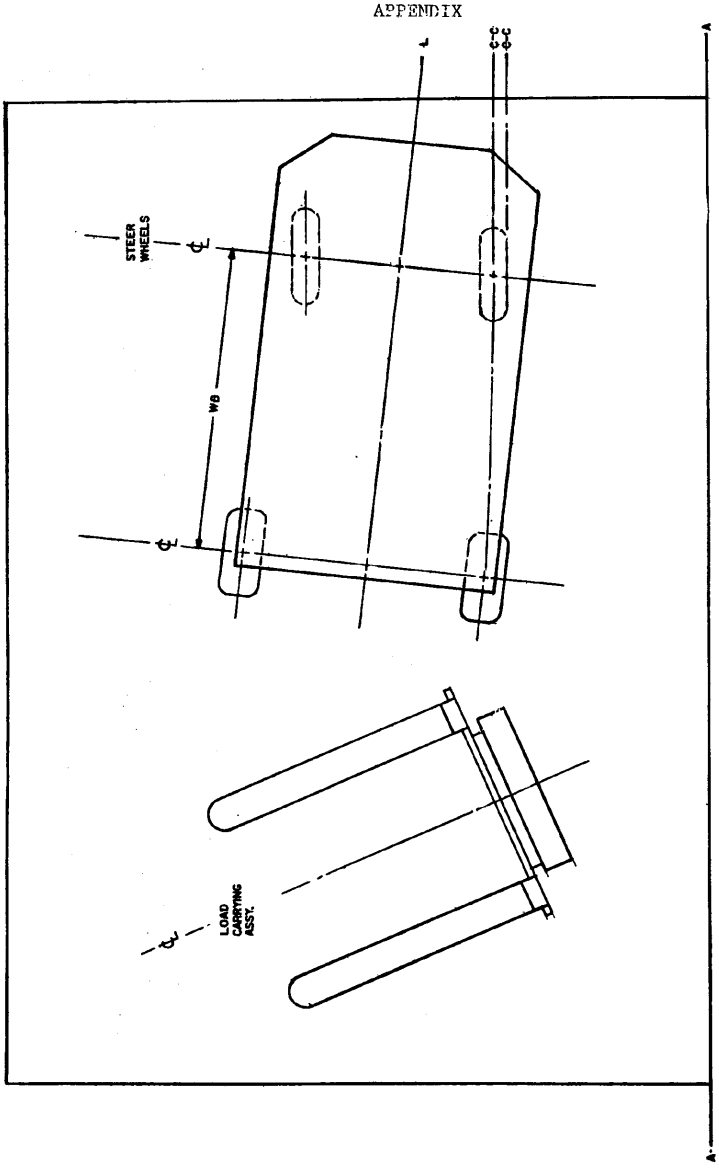


NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

FIGURE A-12. Lateral stability-stacking (L.H). X-3994A
(articulated steer axle)
(counterbalanced front/side loading operation).

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X3646

FIGURE A-13. Lateral stability - steering (non-articulated steer axle) (counterbalanced front/side loading operation).

NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE

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